

Beyond Morin, and deeper into A_7 :

the corank filtration of Thom polynomials,
and the anatomy of Chern positivity at $d \leq 7$

Chern++

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Abstract

We place Rimányi’s conjecture inside a filtration $C_1 \subset C_2 \subset \dots$ of cones of symmetric functions, running from the Chern-monomial cone to the Schur-positive cone. Rimányi’s conjecture says that $\text{Tp}_{A_d} \in C_1$. The natural generalisation is $\text{Tp}_\eta \in C_{\text{corank } \eta}$. We prove the matching lower bound, and we prove, by exact certificates, that the generalisation itself is false. The minimal rank $\rho(\eta)$ is therefore a new invariant, and we tabulate it on the published registry. For Morin singularities we give a sweep algorithm for the chamber series that is 50–100× faster than the previous one. It yields Tp_{A_7} exactly for $\ell \leq 5$ and Tp_{A_6} for $\ell \leq 8$, all positive. We locate where positivity is tight (the plane sector, whose values are Kreweras numbers), and we observe a quantitative strengthening: the negative mass of every packet is at most a fixed fraction of a single term. Every statement is labelled *proved*, *verified* (a finite exact computation) or *conjectured*.

Summary of results

	statement	status
2	$\rho(\eta) \geq \text{support corank} \geq \text{corank } \eta$	proved
3	$\sum \text{Chern coefficients} = [s_{(n)}] \text{Tp}; = 0 \text{ at corank} \geq 2$	proved
4	$\text{Tp}_{I_{2,4}}, \text{Tp}_{III_{2,3}} \notin C_2$; $\rho \neq \text{corank}$	proved
Table 1	ρ on the registry; $\rho - \text{corank}$ reaches 3	verified
§2	Tp_{A_7} ($\ell \leq 5$), Tp_{A_6} ($\ell \leq 8$) exact and positive, $\min C = 1$	verified
8	plane sector: $C(M) = b(M)$ (ballot words) = Kreweras	proved*
9	$C(M) \geq b(M)$, equality iff $\max M \leq 1$ (implies Rimányi)	conjectured
11	$N(M) \leq \kappa_d A_{\beta_{\max}(M)}$ with $\kappa_5 = \kappa_6 = \frac{1}{6}$, $\kappa_7 = \frac{8}{21}$	conjectured
§4	0-Hecke landscape P_5, P_6 (stable), P_7 (on boxes)	verified

* given two geometric inputs (a curvilinear incidence formula and a hyperplane lemma), stated in full in [report/ballot.pdf](#).

1 The corank filtration

Write a Thom polynomial in the relative Chern classes and read c_k as the complete homogeneous symmetric function h_k . Then $s_\nu = \det(c_{\nu_i + j - i})$, and Pragacz–Weber positivity [3] says that every Tp_η is a nonnegative combination of the s_ν .

Definition 1. For $r \geq 1$ let C_r be the convex cone spanned by all products $s_{\nu^1} \cdots s_{\nu^k}$ in which every ν^i has at most r rows. Set $\rho(\eta) = \min\{r : \text{Tp}_\eta \in C_r\}$.

Since $s_{(k)} = c_k$, the cone C_1 is spanned by the Chern monomials; because Chern monomials form a basis, $f \in C_1$ exactly when all its Chern coefficients are nonnegative. At the other end,

$\bigcup_r C_r$ is the Schur-positive cone. Hence:

$$\text{Rimányi's conjecture} \iff \rho(A_d) = 1, \quad \text{Pragacz-Weber} \iff \rho(\eta) < \infty.$$

The filtration is natural from the point of view of residue formulas. The Bérczi–Szenes formula is a residue over $\mathbb{P}(TX)$: after the residue, each chamber monomial becomes a product of one-row Schur functions $c_{\ell+1+\alpha_i}$, and Rimányi's conjecture asks the packet sums to be nonnegative. A formula of the same kind on the Grassmannian $\text{Gr}(r, TX)$ would produce products of Schur functions with at most r rows. So *a corank- r residue formula with nonnegative packet sums would place Tp_η in C_r .*

Theorem 2 (lower bound; proved). *If every Schur component of Tp_η has at least r rows, then $\text{Tp}_\eta \notin C_{r-1}$. In particular $\rho(\eta) \geq \text{corank } \eta$.*

Proof. Let $g = s_{\nu^1} \cdots s_{\nu^k}$ with every $\ell(\nu^i) \leq r-1$. By the Littlewood–Richardson rule, $s_{\nu^1+\dots+\nu^k}$ (row-wise sum) appears in g with coefficient 1, and it has at most $r-1$ rows. All LR coefficients are nonnegative, so in a nonnegative combination of such products nothing cancels it. Every nonzero element of C_{r-1} therefore has a Schur component with at most $r-1$ rows, and Tp_η has none. A singularity of corank r lies in the Thom–Boardman stratum Σ^r , and every Schur component of its Thom polynomial contains the rectangle $(\ell+r)^r$. This is the support property, verified on every registry table: the column s of Table 1 is at least the corank. $\square$

Corollary 3 (proved). *The sum of the Chern coefficients of any Tp_η equals its coefficient on $s_{(n)}$, where $n = \text{codim } \eta$. This is $(d!)^{\ell+1}$ for A_d and 0 for every η of corank ≥ 2 . In particular, no Thom polynomial of corank ≥ 2 is Chern-positive, and each has at least one negative Chern coefficient.*

Proof. $h_\mu = \sum_\lambda K_{\lambda\mu} s_\lambda$, and $K_{(n),\mu} = 1$ for every $\mu \vdash n$. So $[s_{(n)}] \sum_\mu a_\mu h_\mu = \sum_\mu a_\mu$. At corank ≥ 2 every component has at least two rows. For A_d the value $(d!)^{\ell+1}$ is verified: on every published table (tier 14) and on all fifteen new tables of §2. $\square$

The negative Chern coefficients of non-Morin Thom polynomials are thus *forced*. They are not an accident of the data, and they are exactly what Theorem 2 forbids. The natural generalisation of Rimányi's conjecture would be $\rho = \text{corank}$. It is false.

Proposition 4 (proved by certificate). $\text{Tp}_{I_{2,4}} = 16s_{42} + 4s_{33} + 12s_{321} + 5s_{222} + 2s_{2211}$ (at $\ell = 0$) is not in C_2 . Similarly $\text{Tp}_{\text{III}_{2,3}} = 8s_{53} + 4s_{431} + 2s_{332}$ (at $\ell = 1$) is not in C_2 . Both lie in C_3 .

Proof. Since products of Schur functions are Schur-positive, a decomposition can only use generators g with $\text{supp } g \subseteq \text{supp } \text{Tp}$, and there are finitely many. For $I_{2,4}$ the functional $\varphi = [s_{33}] + [s_{222}] - [s_{321}]$ is nonnegative on each of them but $\varphi(\text{Tp}) = 4 + 5 - 12 = -3$. For $\text{III}_{2,3}$ take $\varphi = [s_{332}] - [s_{431}]$: for instance $s_{33}s_2 = s_{53} + s_{431} + s_{332}$, while $\varphi(\text{Tp}) = -2$. The nonnegativity checks, and the C_3 decompositions, are exact rational computations in `chernpp.corank` (tier 14). $\square$

The invariant ρ on the registry. Table 1 gives ρ for the 94 tables decided so far, out of 128 non-Morin tables in the registry. The rest are of codimension 20 to 26 and exceeded the time budget. Three tables ($I_{2,5}$, $\text{III}_{2,6}$ and $\text{III}_{3,5}$ at $\ell = 2$) whose float Farkas vector did not rationalise are now decided by recovering the Farkas vector exactly at an LP vertex (solving the square system on its active support in rationals). Each entry is decided by a linear programme whose answer is re-verified in exact arithmetic, with a decomposition or a Farkas vector as appropriate. The Farkas vectors are strikingly simple: every one found is $\pm \frac{1}{k}$ on a handful of Schur functions. What the table shows:

- $\rho(I_{a,b}) = 2$ exactly when $b \leq a + 1$, and 3 otherwise, for $a \leq b$. Likewise $\rho(\text{III}_{a,b}) = 2$ exactly when $a = b$. In both families ρ is constant in ℓ wherever it was computed.
- ρ is not always constant in ℓ : $\rho(B_4) = 2$ at $\ell = 1$ and 3 at $\ell = 2$.
- $\rho - s$ reaches 3: $B_{5,4}$ at $\ell = 2$ has $s = 2$ and $\rho = 5$. $B_{5,3}$ at $\ell = 1, 2$ has $s = 2$ and $\rho = 4$; C_6 and $C_{5,3}$ have $s = 3$ and $\rho = 4$. Along the B_5 family at $\ell = 2$ the gap runs 0, 0, 2, 3 for $B_{5,1}, \dots, B_{5,4}$.

We have not found an invariant of the local algebra that predicts ρ in general; the multiplicity and the Hilbert function do not. This is the open question the filtration raises.

Remark 5 (a negative result). A sharper variant replaces cones by a basis: $b_\lambda^{(r)} = \prod_k s_{(\lambda_{rk+1}, \dots, \lambda_{rk+r})}$. It is unitriangular against h_λ , equals h_λ at $r = 1$, and so has unique coefficients, as Rimányi's conjecture does. Tp is not $b^{(\text{corank})}$ -positive in general. $I_{2,2}$, $I_{3,3}$, $I_{4,4}$, $\text{III}_{2,2}$ and $\text{III}_{3,3}$ are $b^{(2)}$ -positive wherever tabulated, and $I_{3,4}$ at $\ell = 0$. But $\text{III}_{4,4}$ is not, $I_{2,3}$ fails from $\ell = 2$, and $I_{2,4}$, $I_{4,5}$ and B_4 fail already at the smallest ℓ .

Remark 6 (the shape of C_2). One might hope that C_2 , like C_1 , is generated by a basis. Then membership would be nonnegativity of unique coefficients, and $\rho = 2$ would be a Rimányi-type statement in that basis. The exact extreme rays and facets of C_2 , in the Schur basis of degree n (normaliz, `tools/c2_cone.py`), show that this holds only in low degree:

n	2	3	4	5	6	7	8	9	10
dimension $p(n)$	2	3	5	7	11	15	22	30	42
extreme rays	2	3	5	7	13	17	28	40	61
facets	2	3	5	7	14	18	38	86	598

C_2 is simplicial exactly for $n \leq 5$. At $n = 4$ its extreme rays are s_{22} , s_{11}^2 , s_{31} , $s_2 s_{11}$ and s_4 . From $n = 6$ on it has more extreme rays than its dimension, and the number of facets grows fast. So there is no basis version of C_2 , and ρ is a genuine invariant of the cone. The certificate of Proposition 4 for $I_{2,4}$ lives in degree 6, the first non-simplicial degree.

Table 1: $\rho(\text{Tp}) = \min\{r : \text{Tp} \in C_r\}$ on the non-Morin tables of the registry, by relative dimension ℓ , with the support corank s (every Schur component contains $(\ell + s)^s$). Every entry is decided with an exact certificate; a dash means not in the registry or not yet surveyed.

singularity	s	$\ell = 0$	$\ell = 1$	$\ell = 2$	$\ell = 3$	$\ell = 4$	$\ell = 5$
$I_{2,2}$	2	2	2	2	2	2	2
$I_{2,3}$	2	2	2	2	2	2	—
$I_{2,4}$	2	3	3	3	3	—	—
$I_{2,5}$	2	3	3	3	—	—	—
$I_{2,6}$	2	3	—	—	—	—	—
$I_{2,7}$	2	3	—	—	—	—	—
$I_{3,3}$	2	2	2	2	2	—	—
$I_{3,4}$	2	2	2	2	—	—	—
$I_{3,5}$	2	3	—	—	—	—	—
$I_{3,6}$	2	3	—	—	—	—	—
$I_{4,4}$	2	2	—	—	—	—	—
$I_{4,5}$	2	2	—	—	—	—	—
$\text{III}_{2,2}$	2	—	2	2	2	2	2
$\text{III}_{2,3}$	2	—	3	3	3	3	3
$\text{III}_{2,4}$	2	—	3	3	3	—	—

Table 1: $\rho(\text{Tp}) = \min\{r : \text{Tp} \in C_r\}$ on the non-Morin tables of the registry, by relative dimension ℓ , with the support corank s (every Schur component contains $(\ell + s)^s$). Every entry is decided with an exact certificate; a dash means not in the registry or not yet surveyed.

singularity	s	$\ell = 0$	$\ell = 1$	$\ell = 2$	$\ell = 3$	$\ell = 4$	$\ell = 5$
III _{2,5}	2	—	3	3	—	—	—
III _{2,6}	2	—	3	3	—	—	—
III _{3,3}	2	—	2	2	2	—	—
III _{3,4}	2	—	3	3	—	—	—
III _{3,5}	2	—	3	3	—	—	—
III _{4,4}	2	—	2	2	—	—	—
$(x^2 + y^3, y^4)$	2	2	—	—	—	—	—
$(x^2 + y^4, xy^2)$	2	3	—	—	—	—	—
B ₄	2	—	2	3	3	—	—
B _{5,1}	2	2	2	2	—	—	—
B _{5,2}	2	—	2	2	—	—	—
B _{5,3}	2	—	4	4	—	—	—
B _{5,4}	2	—	—	5	—	—	—
B ₆	2	2	2	2	—	—	—
C ₃	3	—	—	—	3	3	—
C _{4,1}	3	—	—	3	3	—	—
C _{4,2}	3	—	—	3	3	—	—
C _{4,3}	3	—	—	—	4	—	—
C _{5,1}	3	—	3	3	3	—	—
C _{5,2}	3	—	3	3	—	—	—
C _{5,3}	3	—	—	4	—	—	—
C _{5,4}	3	—	3	3	—	—	—
C _{5,5}	3	—	—	4	—	—	—
C _{5,6}	3	—	—	4	—	—	—
C _{5,7}	3	—	—	4	—	—	—
C ₆	3	—	4	4	—	—	—
Σ^3 open (cup)	3	3	—	—	—	—	—

2 New exact data for A_6 and A_7

The sweep. Every monomial of a level- ℓ denominator factor of $F_d = \mathcal{N}_d / \prod_r (1 - f_r)$ contains $x_{\ell-1}$ exactly once. Hence $H = G + fH$ is a recurrence along one axis, and multiplying by $1/(1 - f_r)$ needs one pass over the grid instead of a fixed point. At $d = 7$ this is 34 passes, not ~ 4500 . Values are computed modulo word-sized primes, reconstructed by CRT, and checked against a prime held back from the reconstruction (**chernpp.bboxes**). The sweep agrees cell by cell with the fixed point of **chernpp.chern**, and packet by packet with **chernpp.crt**.

Boxes and packets. Every ballot ordering of M satisfies $\beta \leq \beta_{\max}(M)$, the partial sums of the decreasing ordering. So the level box $\beta_j \leq (d - j)L$ holds every packet with $\min M \geq -L$, and the charge box $\beta_j \leq jp$ holds every packet with $\max M \leq p$. Since $C(M)$ does not depend on ℓ , one level box gives Tp_{A_d} at every $\ell < L$.

Observation 7 (verified). Tp_{A_7} for $\ell \leq 5$ (8033 Chern coefficients) and Tp_{A_6} for $\ell \leq 8$ (12692)

have every coefficient positive, and the minimum is 1. Both agree with every published table they overlap. The A_7 box has 49.6M cells and took seven minutes with four primes. The data are in `results/morin_a{6,7}.json`.

3 Where positivity is tight

For a zero-sum multiset M let $b(M)$ be the number of its *ballot orderings*: distinct orderings whose proper partial sums are nonnegative. Equivalently, $b(M)$ is the packet sum of the pure chain series $\prod_j (1 - x_j)^{-1}$, all of whose coefficients are 1.

Theorem 8 (plane sector; proved, given two geometric inputs). *If $\max M \leq 1$ then $C(M) = b(M)$. Writing $M = (1^{d-b}, 1-s_1, \dots, 1-s_b)$ with $s_i \geq 1$, this is Kreweras' number of noncrossing partitions of $[d]$ with block sizes s_i :*

$$C(M) = b(M) = \frac{d!}{(d-b+1)! \prod_i m_i!},$$

where m_i is the number of blocks of size i .

Proof. Put $n = d + 1$. Four inputs, stated in full in the companion note `report/ballot.pdf` (§2.1), identify $C(M)$ on this sector with the coefficient described below:

- the curvilinear form of the Bérczi–Szenes incidence formula;
- an elementary extraction of Chern coefficients;
- a multiplicity-one hyperplane lemma;
- Briançon's theorem with the Lehn–Sorger description of the punctual Hilbert scheme of the plane.

Concretely, $C(M)$ is the coefficient of an n -cycle σ in $m_\nu(J_1, \dots, J_n)$. Here the $J_k = \sum_{a < k} (a \ k)$ are the Jucys–Murphy elements and ν is the partition formed by the entries of $1 - M$. Expanding the monomial symmetric function, this coefficient counts pairs (e, w) . Here $e = (e_1, \dots, e_n)$ is a rearrangement of $1 - M$ (with $e_1 = 0$, since $J_1 = 0$). And w is a *minimal monotone factorisation* of σ : a product of $n - 1$ transpositions $(a_i \ b_i)$, $a_i < b_i$, with $b_1 \leq \dots \leq b_{n-1}$, exactly e_k of them having $b_i = k$. We show that, for $\sigma = (x \mapsto x + 1)$, each e carries exactly one w when it satisfies the ballot condition $D(j) := j - 1 - \sum_{k \leq j} e_k \geq 0$ for all j , and none otherwise. Reading $\alpha_k = 1 - e_{k+1}$ identifies such e with the ballot orderings of M , which gives $C(M) = b(M)$.

Necessity. The transpositions of a minimal factorisation of an n -cycle form a tree on $[n]$. Those with $b_i \leq j$ form a forest on $[j]$, so there are at most $j - 1$ of them.

Existence and uniqueness, by induction on n . Let the transpositions through n be $(a_1 \ n) \cdots (a_m \ n)$, with $m = e_n$, and write C for their product. The rest multiplies to $F = \sigma C^{-1}$, a permutation of $[n-1]$ with its own minimal monotone factorisation. Since $C^{-1} = (n \ a_1 \ a_2 \ \cdots \ a_m)$, we get $F(x) = x + 1$ for $x \notin \{a_i\}$, $F(a_i) = a_{i+1} + 1$, $F(a_m) = 1$, and $a_1 = \sigma^{-1}(n) = n - 1$. As $FC = \sigma$ is a single cycle and C is an $(m + 1)$ -cycle, each cycle of F contains exactly one a_i . Following $x \mapsto x + 1$ from $a_{i+1} + 1$ we must reach a_i before any other a_j , so $a_1 > a_2 > \dots > a_m$. Hence F is the product of the increasing cycles on the intervals $[1, a_m]$, $[a_m + 1, a_{m-1}]$, $\dots$, $[a_2 + 1, n - 1]$. Minimality on each interval forces $D(a_m) = 0$, $D(a_{m-1}) = 1$, $\dots$, with $D > i$ strictly inside the $(i + 1)$ -st interval. So a_{m-i} is the *last* visit of the walk D to level i . Since D rises by at most 1 per step, these last visits exist and are increasing whenever e is ballot. The cut points are therefore forced and always available. Commuting transpositions on disjoint intervals merge into a unique monotone order, and the induction closes.

The last equality is the cycle lemma. Append -1 to $-M$; exactly one of the n cyclic rotations of any arrangement has all proper partial sums nonnegative, so $b(M) = \frac{1}{n} \binom{n}{d-b+1, m_1, m_2, \dots}$. The count of monotone factorisations by exponent vector is checked exhaustively for $n \leq 7$ (the totals are the Catalan numbers 1, 2, 5, 14, 42, 132), and $C(M) = b(M)$ on every plane-sector packet at $d \leq 7$ (`chernpp.sectors`, tier 12). $\square$

So on the plane sector the Morin chamber series behaves exactly like the pure chain series $\prod_j(1 - x_j)^{-1}$: every negative A_β there cancels to leave one unit per ballot word. The data say this is the floor everywhere.

Conjecture 9 (ballot). *For every zero-sum M , $C(M) \geq b(M)$, with equality exactly when $\max M \leq 1$. Equivalently, $F_d - \prod_j(1 - x_j)^{-1}$ has nonnegative packet sums, vanishing exactly on the plane sector.*

Since the decreasing ordering is always ballot, $b(M) \geq 1$. So Conjecture 9 implies Rimányi’s conjecture, and it explains why the least Chern coefficient is 1 and never 0. It holds on every exact coefficient computed, with no violation and no equality off the plane sector. That is 133,104 packet checks across the tables for $d = 2, \dots, 7$, including every packet of charge ≤ 7 at $d = 7$, all of Tp_{A_7} for $\ell \leq 5$ and Tp_{A_6} for $\ell \leq 12$ (`results/ballot_conjecture.json`). Off the plane sector the ratio $C(M)/b(M)$ is at least 2 throughout. Its minimum is $2, 2, \frac{9}{4}, \frac{12}{5}, \frac{79}{30}, \frac{58}{21}$ for $d = 2, \dots, 7$, attained on the family $(2, 1^{d-4}, -1^{d-2})$ for $d = 5, 7$, with a 0 in place of one 1 at $d = 6$. A naive refinement, weighting each ballot word by $\prod_{\alpha_i > 0} \alpha_i!$ (exact for $(p, -1^p)$, where $C = p!$), fails already at $d = 3$.

A self-contained account, with the evidence table, is in the companion note `report/ballot.pdf`.

Remark 10 (what a proof must explain). In the geometric model (stated in `report/ballot.pdf`, §2.1), $C(M) = \int_{C_{n,r}} m_\nu(E)$ over the closure of curvilinear length- n subschemes of $\mathbb{C}^r$, with the hyperplane lemma lowering r one step at a time. The plane sector is the case where one can descend to $r = 2$, the punctual Hilbert scheme of the plane, where the Jucys–Murphy calculus applies. A packet of charge $\max M = p$ descends only to $r = p + 1$. Conjecture 9 says that the extra dimensions only ever *add*: the integral over the curvilinear locus in $\mathbb{C}^{p+1}$ is at least the ballot count that the plane would give. This is the precise sense in which charge ≥ 2 is where the open problem lives.

Off the plane sector the margin is large. The smallest $C(M)$ with $\max M = 2$ is 12, 17, 27 at $d = 5, 6, 7$, and the layer minima grow quickly with the charge (Table 2).

Anatomy of a packet. Let $N(M)$ be the negative mass of a packet, $\sum_{A_\beta < 0} |A_\beta|$ over its ballot orderings, and $A_{\text{dom}}(M) = A_{\beta_{\max}(M)}$ the coefficient of its decreasing ordering. Table 2 gives, per charge layer, the largest ratio N/A_{dom} .

Conjecture 11 (dominance). *For every d there is $\kappa_d < 1$ with $N(M) \leq \kappa_d A_{\beta_{\max}(M)}$ for every zero-sum M . In particular $C(M) \geq (1 - \kappa_d) A_{\beta_{\max}(M)} > 0$.*

The evidence: $\kappa_5 = \kappa_6 = \frac{1}{6}$ and $\kappa_7 = \frac{8}{21}$ on every packet with $\max M \leq 8, 6, 5$ respectively (exact). The extremal packets are $(1, 1, 0, -1, -1)$, $(1, 1, 1, 0, -1, -2)$ and $(2, 1, 1, 0, -1, -1, -2)$. In every layer the ratio falls by a factor of 3 to 4 per unit of charge, so the conjecture is a statement about small charge. Three remarks keep it honest.

- Individual coefficients are unbounded below: $\min A_\beta = -3622$ in the $d = 7$ layer $p = 5$. This corrects the report’s earlier “least value -2 ”.
- κ_d grows with d . The conjecture is informative only if it stays below 1, and $d = 8$ would test that.
- The single term $A_{\beta_{\max}(M)}$ is the largest in all but 42 of the 3538 packets at $d = 7$. A proof would need a lower bound for this one coefficient of F_d and an upper bound for the negative ones, and no pairing is involved. This is the kind of statement the report’s Assessment §4 asked for.

4 The 0-Hecke landscape

On a packet, let π_i act by $(\pi_i f)(\alpha) = f(\alpha) + f(s_i \alpha)$ if $\alpha_i > \alpha_{i+1}$, by $f(\alpha)$ if $\alpha_i = \alpha_{i+1}$, and by 0 otherwise. The π_i satisfy the 0-Hecke relations (tier 11), and Π_{w_0} puts $C(M)$ on the decreasing

Table 2: Charge layers max $M = p$ (every packet complete). $N(M)$ is the negative mass of the packet and A_{dom} the coefficient of its decreasing ordering $\beta_{\text{max}}(M)$.

d	p	packets	with $A_\beta < 0$	$\min C(M)$	$\min A_\beta$	$\max N(M)/A_{\text{dom}}$
5	1	6	1	1	-1	1/6
5	2	23	3	12	-1	1/20
5	3	54	3	38	-2	1/80
5	4	108	3	24	-3	1/258
5	5	185	3	336	-4	$1.1e - 03$
5	6	297	3	1296	-5	$2.8e - 04$
5	7	441	5	3696	-6	$6.8e - 05$
5	8	632	5	2880	-10	$1.6e - 05$
6	1	10	2	1	-1	1/6
6	2	47	7	17	-3	1/12
6	3	141	18	55	-6	3/103
6	4	333	35	202	-20	3/296
6	5	674	52	120	-70	$3.5e - 03$
6	6	1226	73	2400	-252	$1.2e - 03$
7	1	14	5	1	-4	3/8
7	2	90	43	27	-26	8/21
7	3	331	134	154	-120	43/350
7	4	931	339	326	-516	9/223
7	5	2172	659	1284	-3622	$1.6e - 02$

ordering. The operators and this criterion are from [4, §3.3]. So $P_d = \{w : \Pi_w F_d \geq 0\}$ contains w_0 iff Rimányi's conjecture holds. P_d is an upper set in the left weak order, and its minimal elements are the minimal cancellation units.

Table 3: The 0-Hecke positivity set P_d restricted to the packets complete in the level box $\min M \geq -L$.

d	L	packets	with $A_\beta < 0$	$ P_d $	minimal elements
5	5	377	12	114	4
6	4	532	39	643	11
6	5	1206	76	643	11
7	2	105	39	3998	23
7	3	436	141	3352	49
7	4	1367	374	2792	30
7	5	3539	845	2696	33

- $P_5 = S_5 \setminus \langle s_3, s_4 \rangle$, with minimal elements $s_1, s_2, s_2 s_3, s_2 s_3 s_4$. This is unchanged from $L = 4$ to 5.
- $|P_6| = 643$ with 11 minimal elements, unchanged from $L = 4$ to 5. $s_1 \in P_6$, which is the unpaired-tail and paired inequalities of the τ frame holding at $d = 6$.
- At $d = 7$, $s_1 \notin P_7$, which is the failure of the τ frame at $d = 7$. P_7 still shrinks with the box ($|P_7| = 3998, 3352, 2792, 2696$ for $L = 2, \dots, 5$). The minimal parabolic subsets J with $w_0(J) \in P_7$ are the same at $L = 4$ and 5: $\{1, 2, k\}$ ($k = 3, \dots, 6$) and $\{1, 3, 5\}$. At $d = 5$ they are $\{1\}$ and $\{2\}$, and at $d = 6$ $\{1\}$ and $\{2, 5\}$. The smallest unit of symmetrisation that clears every negative grows from one transposition to three, so no fixed-size pairing

survives.

Exclusions from P_d are proofs, each witnessed by a packet. Memberships hold on the stated boxes only.

5 Stabilisation, and what this means for $\mathcal{Q}_8$

Observation 12 (verified, $d = 5, 6, 7$). $F_d|_{x_{d-1}=0} = F_{d-1}$, cell by cell, on the tested boxes. The corresponding statement for numerators, $[z_d^{e(d)}]\mathcal{Q}_d = (-1)^{e(d)}\mathcal{Q}_{d-1}$ with $e(d) = \lfloor d^2/4 \rfloor - d + 1$, was checked directly at $d = 5$.

The corank filtration changes the case for $\mathcal{Q}_8$. Evidence about generalisations of the conjecture now comes from breadth across coranks, which the registry already supplies. Depth in d adds one Morin data point, and the published Tp_{A_8} and Tp_{A_9} at $\ell = 0$ already lie in C_1 . $\mathcal{Q}_8$ matters for one question raised here: whether $\kappa_8 < 1$ in Conjecture 11. The constraints already available for any candidate $\mathcal{Q}_8$ are its degree (22), its normalisation, the stabilisation $[z_8^9]\mathcal{Q}_8 = -\mathcal{Q}_7$, and agreement with Rimányi’s Tp_{A_8} at $\ell = 0$.

Reproducing

- `tools/morin_tables.py 7 6 results/morin_a7.json` (the A_7 tables).
- `tools/packet_anatomy.py`, `tools/hecke_landscape.py`, and `tools/corank_survey.py` for ρ .
- `tools/render_d7_tables.py report/tables` regenerates every table here.

Tiers 10–14 of the test suite pin each module.

References

- [1] L. Bérczi, A. Szenes, Thom polynomials of Morin singularities, *Ann. of Math.* 175 (2012) 567–629.
- [2] R. Rimányi, Thom polynomials, symmetries and incidences of singularities, *Invent. Math.* 143 (2001) 499–521.
- [3] P. Pragacz, A. Weber, Positivity of Schur function expansions of Thom polynomials, *Fund. Math.* 195 (2007) 85–100.
- [4] Rimányi positivity: findings summary (29 July 2026), unrefereed, `papers/rimanyi_positivity_findings_summary.pdf`.